Quantum Electronics & Qubits
Exercise Sheet 2 - Solutions

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Solutions generated by AI model

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Exercise 1: Non-additivity of QPC Resistances in Series

MFIGURE

Figure Placeholder: Figure 1

Sketch of the two constrictions acting as QPCs, and measurements of the 2-terminal magnetoconductance for different values of the gate voltage V_{gate} .

(Refer to the original exercise sheet for the actual figure)

We would expect the conductance of two quantum point contacts (QPCs) in series to be half that of a single QPC: $G = M\left(2e^2/h\right) \times 0.5$. However, when the two QPCs are sufficiently close together then the electrons emerging from the first constriction do not have the chance to spread out before they reach the second constriction. Consequently, the conductance can be the same as that of a single QPC. However, this is only true in zero magnetic field, because in the presence of a magnetic field the electrons get deflected!

1. Considering the Kirchoff law that relates the current to conductivity, derive the expression for the total conductivity, using the Landauer-Büttiker formula:

$$\left(\frac{h}{2e^2}\right) I_i = (N_i - R_{ii}) V_i - \sum_{j\neq i} T_{ij} V_j,$$
$$N_i - R_{ii} = \sum_{j\neq i} T_{ji}.$$

with T_{ij} the transmission probability from the j th to the i th contact. Let's assume that a current I flows from the source (V_1) to the drain ($V_3 = 0$), and that inelastic scattering has equilibrated the 2DEG far to the left and right of the point contacts giving rise to the potentials V_2 and V_4, respectively. Consider the general case with external magnetic field (i.e. $T_R \neq T_L$):

$$T_{13} = T_{31} = T_F$$
$$T_{42} = T_{24} = T'_F$$
$$T_{21} = T_{32} = T_{43} = T_{14} = T_R$$
$$T_{41} = T_{12} = T_{23} = T_{34} = T_L$$

Solution to 1.1

Let $g_0 = \frac{2e^2}{h}$. The Landauer-Büttiker formula can be written as:

$$\frac{I_i}{g_0} = \left(\sum_{k \neq i} T_{ki} \right) V_i - \sum_{j \neq i} T_{ij} V_j$$

We are given:

- Current I flows from source (contact 1) to drain (contact 3). So, $I_1 = I$ and $I_3 = -I$.
- Contact 3 is grounded: $V_3 = 0$.
- Contacts 2 and 4 are voltage probes where no net current flows: $I_2 = 0, I_4 = 0$.

The transmission probabilities are: $T_{13} = T_{31} = T_F$ (Forward transmission $1 \leftrightarrow 3$) $T_{42} = T_{24} = T'_F$ (Transmission between cavity probes $2 \leftrightarrow 4$) $T_{21} = T_{32} = T_{43} = T_{14} = T_R$ (Right- transmissions, e.g., $1 \to 2$ related T_{21}, etc.) $T_{41} = T_{12} = T_{23} = T_{34} = T_L$ (Left- transmissions)

Let's write the equations for each contact:

1. **For contact 1 ($I_1 = I$):**

$$\frac{I}{g_0} = (T_{21} + T_{31} + T_{41})V_1 - (T_{12}V_2 + T_{13}V_3 + T_{14}V_4)$$

Substituting $V_3 = 0$ and the given T-coefficients:

$$\frac{I}{g_0} = (T_R + T_F + T_L)V_1 - (T_L V_2 + T_R V_4) \quad \text{(Eq. A)}$$

2. **For contact 2 ($I_2 = 0$):**

$$0 = (T_{12} + T_{32} + T_{42})V_2 - (T_{21}V_1 + T_{23}V_3 + T_{24}V_4)$$

Substituting $V_3 = 0$ and T-coefficients:

$$0 = (T_L + T_R + T'_F)V_2 - (T_R V_1 + T_L \cdot 0 + T'_F V_4)$$

$$(T_L + T_R + T'_F)V_2 = T_R V_1 + T'_F V_4 \quad \text{(Eq. B)}$$

3. **For contact 3 ($I_3 = -I$):**

$$\frac{-I}{g_0} = (T_{13} + T_{23} + T_{43})V_3 - (T_{31}V_1 + T_{32}V_2 + T_{34}V_4)$$

Substituting $V_3 = 0$ and T-coefficients:

$$\frac{-I}{g_0} = 0 - (T_F V_1 + T_R V_2 + T_L V_4)$$

$$\frac{I}{g_0} = T_F V_1 + T_R V_2 + T_L V_4 \quad \text{(Eq. C)}$$

This equation directly gives an expression for I/g_0 if V_2 and V_4 are known in terms of V_1.

4. **For contact 4 ($I_4 = 0$):**

$$0 = (T_{14} + T_{24} + T_{34})V_4 - (T_{41}V_1 + T_{42}V_2 + T_{43}V_3)$$

Substituting $V_3 = 0$ and T-coefficients:

$$0 = (T_R + T'_F + T_L)V_4 - (T_L V_1 + T'_F V_2 + T_R \cdot 0)$$

$$(T_R + T'_F + T_L)V_4 = T_L V_1 + T'_F V_2 \quad \text{(Eq. D)}$$

Let $S_V = T_L + T_R + T'_F$. Equations (B) and (D) become: $S_V V_2 = T_R V_1 + T'_F V_4$ $S_V V_4 = T_L V_1 + T'_F V_2$

Equating I/g_0 from (Eq. A) and (Eq. C): $(T_R + T_F + T_L)V_1 - (T_L V_2 + T_R V_4) = T_F V_1 + T_R V_2 + T_L V_4$ $(T_R + T_L)V_1 = (T_L + T_R)V_2 + (T_R + T_L)V_4$ If $T_R + T_L \neq 0$, then $V_1 = V_2 + V_4$. Let's verify this. Solving for V_2, V_4 from the system: $V_2 = \frac{S_V T_R + T'_F T_L}{S_V^2 - (T'_F)^2} V_1$ and $V_4 = \frac{S_V T_L + T'_F T_R}{S_V^2 - (T'_F)^2} V_1$. $V_2 + V_4 = \frac{(S_V T_R + T'_F T_L) + (S_V T_L + T'_F T_R)}{S_V^2 - (T'_F)^2} V_1 = \frac{S_V (T_R + T_L) + T'_F (T_L + T_R)}{(S_V - T'_F)(S_V + T'_F)} V_1$ $V_2 + V_4 = \frac{(S_V + T'_F)(T_R + T_L)}{(S_V - T'_F)(S_V + T'_F)} V_1 = \frac{T_R + T_L}{S_V - T'_F} V_1$. Since $S_V - T'_F = (T_L + T_R + T'_F) - T'_F = T_L + T_R$, we confirm $V_2 + V_4 = V_1$ (if $T_L + T_R \neq 0$).

Using $V_4 = V_1 - V_2$ in (Eq. B): $S_V V_2 = T_R V_1 + T'_F(V_1 - V_2) = (T_R + T'_F)V_1 - T'_F V_2$ $(S_V + T'_F)V_2 = (T_R + T'_F)V_1 \Rightarrow V_2 = \frac{T_R + T'_F}{S_V + T'_F} V_1$. Then $V_4 = V_1 - V_2 = \left(1 - \frac{T_R + T'_F}{S_V + T'_F}\right) V_1 = \frac{S_V - T_R}{S_V + T'_F} V_1 = \frac{T_L + T'_F}{S_V + T'_F} V_1$. Substitute V_2, V_4 into (Eq. C) for I/g_0: $\frac{I}{g_0} = T_F V_1 + T_R \left(\frac{T_R + T'_F}{S_V + T'_F}\right) V_1 + T_L \left(\frac{T_L + T'_F}{S_V + T'_F}\right) V_1$. The total conductance $G = I/V_1$: $G = g_0 \left[T_F + \frac{T_R(T_R + T'_F) + T_L(T_L + T'_F)}{S_V + T'_F}\right]$ $G = g_0 \left[T_F + \frac{T_R^2 + T_R T'_F + T_L^2 + T_L T'_F}{T_L + T_R + T'_F + T'_F}\right]$ $\mathbf{G = \frac{2e^2}{h}\left[T_F + \frac{T_R^2 + T_L^2 + T'_F(T_R + T_L)}{T_L + T_R + 2T'_F}\right]}$. This is the general expression for the total conductivity. This derivation holds if $S_V + T'_F = T_L + T_R + 2T'_F \neq 0$. If $T_R + T_L = 0$, then $V_1 = V_2 + V_4$ might not hold. From (Eq A) and (Eq C), $(T_R + T_L)V_1 = (T_L + T_R)V_2 + (T_R + T_L)V_4$. If $T_R + T_L = 0$, then $0 = 0$, so $V_1 = V_2 + V_4$ is not necessarily true. If $T_R = T_L = 0$: The formula gives $G = g_0 \left[T_F + \frac{0}{2T'_F}\right] = g_0 T_F$ (if $T'_F \neq 0$). If $T_R = T_L = T'_F = 0$: The formula is undefined. However, from (Eq. B) and (Eq. D), $0 = 0$. From (Eq. C), $I/g_0 = T_F V_1$, so $G = g_0 T_F$. The formula should gracefully handle this. If $T_L, T_R \to 0$, the fraction becomes $\frac{T'_F(T_R + T_L)}{2T'_F} = (T_R + T_L)/2 \to 0$. So it is consistent.

2. For a large separation of the two QPCs, direct transmission is suppressed $T_F \approx 0$. Show that the Ohmic addition of resistance applies in zero magnetic field.

Solution to 1.2

In zero magnetic field ($B = 0$), there is no preferred direction for deflection, so $T_R = T_L$. Let's call this T. The transmission coefficients T_{ij} are total transmissions (summed over modes). $T_{21} = T_R = T$: from source (1) to cavity left (2) via QPC1. $T_{41} = T_L = T$: from source (1) to cavity right (4) via QPC1. So the total transmission of QPC1 is $T_{QPC1} = T_{21} + T_{41} = T + T = 2T$. Similarly for QPC2, its transmission is $T_{QPC2} = T_{32} + T_{34} = T + T = 2T$. Let $T_{QPC} = 2T$. The conductance of a single QPC is $G_{QPC} = g_0 T_{QPC} = g_0(2T)$. Ohmic addition of two identical QPCs in series means $R_{total} = R_{QPC} + R_{QPC} = 2R_{QPC}$. So $G_{total} = \frac{1}{2R_{QPC}} = \frac{G_{QPC}}{2} = \frac{g_0(2T)}{2} = g_0 T$. We need to show that our general formula for G reduces to $g_0 T$.

The general formula for G is $g_0 \left[T_F + \frac{T_R^2 + T_L^2 + T'_F(T_R + T_L)}{T_L + T_R + 2T'_F}\right]$. Substitute $T_F \approx 0$ and $T_R = T_L = T$: $G = g_0 \left[0 + \frac{T^2 + T^2 + T'_F(T + T)}{T + T + 2T'_F}\right] = g_0 \left[\frac{2T^2 + 2T'_F T}{2T + 2T'_F}\right]$. $G = g_0 \left[\frac{2T(T + T'_F)}{2(T + T'_F)}\right]$. If $T + T'_F \neq 0$, then $G = g_0 T$. This matches the $G_{QPC}/2$ result, confirming Ohmic addition.

For large separation, electrons equilibrate in the cavity between the QPCs. This means the cavity acts as a reservoir, and probes 2 and 4 should be at the same potential, $V_2 = V_4$. This implies that T'_F (transmission between probe 2 and 4) is effectively very large ($T'_F \to \infty$), ensuring equilibration. If $T'_F \to \infty$: $G = g_0 \left[\frac{2T^2 + 2T'_F T}{2T + 2T'_F}\right] = g_0 \lim_{T'_F \to \infty} \left[\frac{T'_F(2T^2/T'_F + 2T)}{T'_F(2T/T'_F + 2)}\right] = g_0 \frac{2T}{2} = g_0 T$. This confirms the result. The assumption of large separation implies equilibration in the cavity, which mathematically means T'_F is large compared to T.

3. Show that for very short separations in zero magnetic field, the conductivity is indeed the same as if there were only one QPC.

Solution to 1.3

For very short separations and $B=0$: Electrons emerging from the first QPC do not scatter or spread out before reaching the second QPC. They pass ballistically from QPC1 to QPC2. This suggests that the dominant transmission is the direct one from source (1) to drain (3), $T_F = T_{13}$. The two QPCs effectively merge into a single scatterer. The transmission of this single effective QPC is T_F. So $G_{singleQPC} = g_0 T_F$. The transmissions into the side probes (T_R, T_L) become negligible because electrons are channelled directly from 1 to 3. So $T_R = T_L = T' \approx 0$. At $B=0$, $T_R = T_L = T'$. The general formula: $G = g_0\left[T_F + \frac{2(T')^2+T'_F(2T')}{2T'+2T'_F}\right] = g_0\left[T_F + \frac{2T'(T'+T'_F)}{2(T'+T'_F)}\right]$. If $T'+T'_F \neq 0$, $G = g_0(T_F+T')$. With $T' \approx 0$: $G \approx g_0 T_F$. If T_F represents the transmission of a single QPC (T_{QPC}), then $G = g_0 T_{QPC}$. This means the conductance is the same as that of a single QPC.

4. Now, we turn on the magnetic field! The Lorentz force will deflect the electrons, increasing T_R at the expense of T_F and T_L. In a sufficiently strong magnetic field (i.e. $2l_c < L$ with $l_c \equiv \hbar k_F/eB$ the cyclotron radius in the region between the QPCs), T_F and T_L become negligibly small compared to T_R, so that only indirect transmission remains. Moreover, in a strong magnetic field backscattering in the region between the QPC is suppressed $R_{RR} \approx R_{LL} \approx 0$. Show that the expression for the total conductivity simplifies to:

$$G = \frac{2e^2}{h}\left(\frac{2}{N_{\mathrm{QPC}}} - \frac{1}{N_{\mathrm{2DEG}}}\right)^{-1}$$

with N_{QPC} and N_{2DEG} the number of modes in the QPC (i.e. region 1 and 3) and 2 DEG (i.e. region 2 and 4), respectively.

Solution to 1.4

Given conditions for strong magnetic field:

- $T_F \approx 0$
- $T_L \approx 0$
- $R_{ii} \approx 0$ for $i = 1,2,3,4$. (Assuming R_{RR}, R_{LL} implies all R_{ii} for relevant contacts).

The general formula for conductance is $G = g_0\left[T_F + \frac{T_R^2+T_L^2+T'_F(T_R+T_L)}{T_L+T_R+2T'_F}\right]$. Substitute $T_F \approx 0$ and $T_L \approx 0$: $G = g_0\left[0 + \frac{T_R^2+0+T'_F(T_R+0)}{0+T_R+2T'_F}\right] = g_0\left[\frac{T_R^2+T'_F T_R}{T_R+2T'_F}\right] = g_0\frac{T_R(T_R+T'_F)}{T_R+2T'_F}$.

Now use the relation $N_i - R_{ii} = \sum_{j\neq i} T_{ji}$. With $R_{ii} \approx 0$, this becomes $N_i = \sum_{j\neq i} T_{ji}$. N_{QPC} is the number of modes in QPC regions (1 and 3). So $N_1 = N_3 = N_{QPC}$. N_{2DEG} is the number of modes in 2DEG regions (2 and 4). So $N_2 = N_4 = N_{2DEG}$.

- For contact 1 ($N_1 = N_{QPC}$): $N_{QPC} = T_{21} + T_{31} + T_{41}$. Given T-coefficients: $T_{21} = T_R$, $T_{31} = T_F$, $T_{41} = T_L$. With $T_F \approx 0$ and $T_L \approx 0$: $N_{QPC} = T_R + 0 + 0 \Rightarrow T_R = N_{QPC}$.
- For contact 2 ($N_2 = N_{2DEG}$): $N_{2DEG} = T_{12} + T_{32} + T_{42}$. Given T-coefficients: $T_{12} = T_L$, $T_{32} = T_R$, $T_{42} = T'_F$. With $T_L \approx 0$: $N_{2DEG} = 0 + T_R + T'_F \Rightarrow N_{2DEG} = T_R + T'_F$.

We have $T_R = N_{QPC}$ and $T_R + T'_F = N_{2DEG}$. From these, $T'_F = N_{2DEG} - T_R = N_{2DEG} - N_{QPC}$. (Note: For T'_F to be a non-negative transmission probability, we must have $N_{2DEG} \geq N_{QPC}$.)

Substitute these into the expression for G: $G = g_0 \frac{N_{QPC}(N_{2DEG})}{N_{QPC}+2(N_{2DEG}-N_{QPC})}$. $G = g_0 \frac{N_{QPC}N_{2DEG}}{N_{QPC}+2N_{2DEG}-2N_{QPC}} = g_0 \frac{N_{QPC}N_{2DEG}}{2N_{2DEG}-N_{QPC}}$. To match the target expression, rewrite as: $G = g_0 \left(\frac{2N_{2DEG}-N_{QPC}}{N_{QPC}N_{2DEG}}\right)^{-1}$. $G = g_0 \left(\frac{2N_{2DEG}}{N_{QPC}N_{2DEG}} - \frac{N_{QPC}}{N_{QPC}N_{2DEG}}\right.$ $\mathbf{G} = \mathbf{\frac{2e^2}{h}} \left(\mathbf{\frac{2}{N_{QPC}}} - \mathbf{\frac{1}{N_{2DEG}}}\right)^{\mathbf{-1}}$. This matches the required expression. This is valid if $2N_{2DEG}-N_{QPC} > 0$, i.e. $N_{2DEG} > N_{QPC}/2$.

5. In such large magnetic fields ($2l_c < L$), the 2DEG forms Landau levels and $N_{2\ \mathrm{DEG}}$ actually equals the number of occupied Landau levels, i.e. $N_{\mathrm{2DEG}} = \frac{1}{2}k_F l_c$. The QPC, on the other hand, will only form Landau levels once $2l_c < W_{\mathrm{QPC}}$. Can you use this knowledge to explain the so-called 'camel back' in the graph of conductivity vs. magnetic field (Fig.1)? Why does G first increase and then decrease? What happens near the peak?

Solution to 1.5

The conductance is $G = g_0 \left(\frac{2}{N_{\mathrm{QPC}}} - \frac{1}{N_{\mathrm{2DEG}}}\right)^{-1}$. We are given $N_{2DEG} = \frac{1}{2}k_F l_c$. Since $l_c = \hbar k_F/(eB)$, $N_{2DEG} = \frac{1}{2}k_F \frac{\hbar k_F}{eB} = \frac{\hbar k_F^2}{2eB}$. This means $N_{2DEG} \propto 1/B$. As the magnetic field B increases, N_{2DEG} decreases.

The behavior of G with B can be understood by considering two regimes for N_{QPC}:

1. **Low to moderate B field (where $2l_c > W_{QPC}$ but $2l_c < L$ is already met):** In this regime, the cyclotron radius l_c is still larger than half the QPC width $W_{QPC}/2$. Landau levels are not yet well-formed *within* the QPC. So, N_{QPC} is primarily determined by the electrostatic confinement due to V_{gate} and W_{QPC}, and is approximately constant with B. As B increases:
   - N_{QPC} is constant.
   - $N_{2DEG} \propto 1/B$ decreases.
   - Therefore, $1/N_{2DEG} \propto B$ increases.
   - The term in the denominator $(2/N_{QPC} - 1/N_{2DEG})$ decreases (since $1/N_{2DEG}$ is subtracted).
   - Consequently, G increases. This explains the initial rise of conductance in Fig.1.

2. **High B field (where $2l_c < W_{QPC}$):** In this regime, l_c is small enough that Landau levels also form within the QPC. N_{QPC} now also becomes dependent on B, behaving similarly to N_{2DEG}, i.e., $N_{QPC}(B)$ also represents the number of Landau levels that can pass through the constriction. So $N_{QPC}(B)$ generally decreases as B increases (typically in steps). As B increases further:
   - $N_{QPC}(B)$ decreases. So $2/N_{QPC}(B)$ increases.
   - $N_{2DEG}(B)$ continues to decrease. So $1/N_{2DEG}(B)$ increases.

   The term $(2/N_{QPC}(B) - 1/N_{2DEG}(B))$: If both N_{QPC} and N_{2DEG} decrease with B (e.g. $N \sim E_F/(\hbar\omega_c)$), then $1/N_{QPC} \propto B$ and $1/N_{2DEG} \propto B$. Let $N_{QPC}(B) \approx C_1/B$ and $N_{2DEG}(B) \approx C_2/B$ (as a smooth approximation). Then $G \approx g_0(2B/C_1 - B/C_2)^{-1} = g_0(B(2/C_1 - 1/C_2))^{-1}$. If $(2/C_1 - 1/C_2)$ is a positive constant (which depends on k_F and W_{QPC} vs k_F and L), then $G \propto 1/B$. This means G decreases as B increases. This explains the fall of conductance after the peak.

Near the peak: The transition between these two regimes occurs when $2l_c \approx W_{QPC}$. This is where N_{QPC} starts to become significantly B-dependent. The peak in G occurs roughly at this magnetic field. The exact peak condition $dG/dB = 0$ would involve the points where N_{QPC} and N_{2DEG} change values (they are step-like functions of B). The formula also implies a resonance when $N_{2DEG} \approx N_{QPC}/2$, where G would diverge. This is usually smoothed out by scattering or temperature effects. The actual peak is likely a complex interplay of $N_{QPC}(B)$ and $N_{2DEG}(B)$ changing. The "camel back" shape arises from this non-monotonic behavior: G first increases due to N_{2DEG} decreasing while N_{QPC} is constant, then decreases as N_{QPC} also starts to decrease and its change dominates the expression for G.

The gate voltage V_{gate} controls W_{QPC} and thus the initial N_{QPC}. A less negative V_{gate} means a wider W_{QPC} and a larger $N_{QPC}(B=0)$. The condition $2l_c \approx W_{QPC}$ implies $B_{peak} \approx 2\hbar k_F/(eW_{QPC})$. So a larger W_{QPC} (higher V_{gate}) leads to a smaller B_{peak}. This trend is visible in Fig.1: curves for higher V_{gate} (e.g., -1.8V) peak at lower B fields than curves for lower V_{gate} (e.g., -2.5V).

Exercise 2: Interferences in a Ring

1. In rings patterned in a high-mobility GaAs-AlGaAs 2DEG, Timp et al. (Surf. Sci. 196, 68 (1988)) observed the oscillations presented in Fig.2. Estimate the period of the oscillations and deduce the diameter of the ring.

Figure Placeholder: Figure 2

MFIGURE *Aharonov-Bohm oscillations in a GaAs/AlGaAs ring*

(Refer to the original exercise sheet for the actual figure)

Solution to 2.1

Aharonov-Bohm oscillations occur with a period ΔB such that the change in magnetic flux through the loop area A is one flux quantum $\Phi_0 = h/e$. $\Delta\Phi = A \cdot \Delta B = \Phi_0$. The area of a ring with diameter d is $A = \pi(d/2)^2 = \pi d^2/4$. So, $d = \sqrt{4A/\pi} = \sqrt{4\Phi_0/(\pi\Delta B)}$. The value of the flux quantum is $\Phi_0 = h/e \approx 4.1357 \times 10^{-15}$ T m^2.

To estimate ΔB from Figure 2: Let's identify the positions of minima (or maxima). Minima appear easier to locate. Approximate positions of minima from the graph: $B_{min1} \approx 0.008$ T $B_{min2} \approx 0.028$ T $B_{min3} \approx 0.049$ T $B_{min4} \approx 0.070$ T $B_{min5} \approx 0.091$ T

Calculating periods between successive minima: $\Delta B_1 = 0.028 - 0.008 = 0.020$ T $\Delta B_2 = 0.049 - 0.028 = 0.021$ T $\Delta B_3 = 0.070 - 0.049 = 0.021$ T $\Delta B_4 = 0.091 - 0.070 = 0.021$ T The average period is $\Delta B \approx (0.020 + 0.021 + 0.021 + 0.021)/4 = 0.02075 \text{ T} \approx 0.021$ T.

Now, calculate the area A: $A = \frac{\Phi_0}{\Delta B} = \frac{4.1357\times10^{-15} \text{ T m}^2}{0.02075 \text{ T}} \approx 1.9931 \times 10^{-13}$ m^2.

Finally, deduce the diameter d: $d = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4\times1.9931\times10^{-13} \text{ m}^2}{\pi}} \approx \sqrt{2.5375 \times 10^{-13} \text{ m}^2}$. $d \approx 5.037 \times 10^{-7}$ m = **503.7** nm.

The paper by Timp et al. mentions rings with average radii of 230 nm (diameter 460 nm) and 340 nm (diameter 680 nm). Our estimated diameter of $\sim$ 504 nm is reasonably close to the 460 nm ring, considering visual estimation from the graph. If the ring was the one with $d = 460$ nm, the expected $\Delta B = \Phi_0/(\pi(230 \times 10^{-9})^2) \approx 0.0249$ T. If $d = 504$ nm, $\Delta B = 0.02075$ T.

2. Suppose we would use a thick ring such that the inner diameter is 1μ m, and the outer one 2μ m. Do you expect to see Aharonov-Bohm oscillations in that case? To what analogous phenomenon in optics can you relate that?

Solution to 2.2

For a thick ring: Inner diameter $d_{in} = 1\mu$m, so inner radius $r_{in} = 0.5\mu$m. Outer diameter $d_{out} = 2\mu$m, so outer radius $r_{out} = 1\mu$m. The width of the ring arms is $w = r_{out} - r_{in} = 1\mu\text{m} - 0.5\mu\text{m} = 0.5\mu\text{m} = 500$ nm.

The mean radius of the ring is $r_{mean} = (r_{in} + r_{out})/2 = (0.5 + 1)/2 = 0.75\mu$m. The mean diameter is $d_{mean} = 1.5\mu$m.

Conditions for observing Aharonov-Bohm (AB) oscillations:

1. **Phase coherence:** The phase coherence length L_ϕ must be larger than the circumference of the ring $C = \pi d_{mean}$. $C = \pi \times 1.5\mu\text{m} \approx 4.71\mu$m. In high-mobility GaAs-AlGaAs 2DEGs at low temperatures, L_ϕ can be several micrometers to tens of micrometers. So, this condition can likely be met.

2. **Well-defined area:** The width of the ring arms w should ideally be much smaller than the mean radius r_{mean} ($w \ll r_{mean}$). In this case, $w = 0.5\mu$m and $r_{mean} = 0.75\mu$m. The ratio $w/r_{mean} = 0.5/0.75 = 2/3$. This is not small; w is comparable to r_{mean}.

If w is large, electrons can take paths along different radii within the ring arm. Each path encloses a slightly different area, leading to a range of ΔB values for a full oscillation. $A_{in} = \pi r_{in}^2 = \pi(0.5 \times 10^{-6}\text{ m})^2 \approx 0.785 \times 10^{-12}\text{ m}^2$. $\Delta B_{in} = \Phi_0/A_{in} \approx (4.1357 \times 10^{-15})/(0.785 \times 10^{-12}) \approx 0.00527$ T. $A_{out} = \pi r_{out}^2 = \pi(1 \times 10^{-6}\text{ m})^2 \approx 3.142 \times 10^{-12}\text{ m}^2$. $\Delta B_{out} = \Phi_0/A_{out} \approx (4.1357 \times 10^{-15})/(3.142 \times 10^{-12}) \approx 0.00132$ T. The superposition of oscillations with different periods (ranging from 1.32 mT to 5.27 mT) will cause the AB oscillations to be significantly smeared out, and possibly become unobservable. This is an averaging effect.

Expectation: We expect to see **weak or strongly suppressed Aharonov-Bohm oscillations**. The amplitude of the oscillations would be much smaller than for a narrow ring.

Analogous phenomenon in optics: This is analogous to interference phenomena where the path difference is not well-defined. For example:

- **Young's double-slit experiment with wide slits:** If the slits are very wide, each slit acts as an array of point sources. Interference patterns from different parts of the slits (e.g., inner edge vs. outer edge of each slit) will have slightly different fringe spacings or will be shifted. The overall observed pattern becomes blurred, and contrast is reduced.

- **Interference from an extended source:** If the light source in an interference experiment is not a point source but is extended, it's like having many point sources contributing to the pattern. If the source is too large, the fringes from different parts of the source overlap and wash out.

In both cases, the lack of a sharply defined geometric parameter (slit separation, path length difference determined by precise area) leads to a smearing of the interference pattern.

3. In Rößler et al. Nanoletters 23, 7, 2846-2853 (2023), Aharonov-Bohm-like oscillations have been reported in a (Bi,Sb)Te nanowire (see Fig.3). By analysing the period of the oscillations, explain how the A-B oscillations are measured. How is the external magnetic field pointing? What area would you expect from the oscillation period? Can you explain why this expected area might differ slightly from the reported dimensions of the nanowire?

Figure Placeholder: Figure 3

MFIGURE *Aharonov-Bohm oscillations in a* $(\mathrm{Bi},\mathrm{Sb})\mathrm{Te}$ *nanowire.*

(Refer to the original exercise sheet for the actual figure)

Solution to 2.3

Analysis of oscillation period: Figure 3 shows magnetoresistance R_{xx} vs. magnetic field B. We need to estimate the period ΔB. Let's identify minima in the oscillations: $B_{min} \approx -2.5$ T, -1.7 T, -0.9 T, -0.1 T, 0.7 T, 1.5 T, The periods between consecutive minima are: $|-1.7-(-2.5)| = 0.8$ T $|-0.9-(-1.7)| = 0.8$ T $|-0.1-(-0.9)| = 0.8$ T $|0.7-(-0.1)| = 0.8$ T $|1.5-0.7| = 0.8$ T $|2.3-1.5| = 0.8$ T $|3.1-2.3| = 0.8$ T The period $\Delta B \approx 0.8$ T. The cited paper (Rößler et al.) reports $\Delta B = 0.75(5)$ T for similar measurements, which is consistent with this estimation. Let's use $\Delta B = 0.75$ T for consistency with the likely source value.

How AB oscillations are measured and B-field direction: AB oscillations are measured by tracking a phase-sensitive property (like conductance or resistance) as a function of magnetic flux. In this case, R_{xx} (longitudinal resistance) is measured while sweeping an external magnetic field B. For a nanowire, especially one made of a topological insulator like (Bi,Sb)Te which hosts surface states, coherent electron paths can encircle the nanowire's circumference. If the magnetic field is applied **parallel to the axis of the nanowire**, these surface state paths will enclose a magnetic flux $\Phi = B \cdot A_{cross}$, where A_{cross} is the cross-sectional area of the nanowire. The oscillations observed are then due to the Aharonov-Bohm effect for these circumferential paths.

Expected area from oscillation period: The oscillations are typically h/e for AB effect (though $h/2e$ can occur, e.g. AAS oscillations). For surface states on TI nanowires, h/e oscillations are expected. $A = \frac{\Phi_0}{\Delta B} = \frac{h/e}{\Delta B}$. Using $\Phi_0 = 4.1357 \times 10^{-15}$ T m^2 and $\Delta B = 0.75$ T: $A = \frac{4.1357 \times 10^{-15} \text{ T m}^2}{0.75 \text{ T}} \approx 5.514 \times 10^{-15}$ m^2.

Comparison with reported dimensions: The nanowire used for measurements has width $w = 150$ nm and thickness $t = 30$ nm. The geometric cross-sectional area (assuming rectangular) is $A_{geom} = w \times t = (150 \times 10^{-9} \text{ m}) \times (30 \times 10^{-9} \text{ m}) = 4500 \times 10^{-18} \text{ m}^2 = 4.5 \times 10^{-15} \text{ m}^2$. The area calculated from ΔB ($A \approx 5.51 \times 10^{-15}$ m^2) is slightly larger than the geometric area ($A_{geom} = 4.5 \times 10^{-15}$ m^2). The difference is about $(5.51 - 4.5)/4.5 \approx 22\%$. The paper considers this "good agreement".

Reasons for discrepancy between expected and geometric area:

1. **Effective electron path area:** The charge carriers (especially in surface states) do not flow precisely on the geometric boundary. The wavefunction of these states has a finite extent and can penetrate into the bulk or extend slightly outside, making the effective area enclosed by the current paths different from the geometric cross-section. For topological surface states, the current path is on the surface, but "surface" isn't an infinitely thin layer.

2. **Actual vs. nominal dimensions:** The reported dimensions (150 nm x 30 nm) are nominal or based on characterization (like SEM). The actual dimensions of the conducting part of the nanowire might vary along its length or differ from the average measured values. SEM typically measures the physical size, not necessarily the "electrical" size.

3. **Cross-sectional shape:** The nanowire cross-section might not be perfectly rectangular. SEM images (like Fig 3a for a 120nm wire) often show deviations (e.g., trapezoidal, rounded corners, or even hexagonal for some nanowires). A different shape would lead to a different area for the same nominal width and thickness. For example, if the corners are rounded, the actual area might be less than $w \times t$. However, the effective path for circumferential current might be larger if the states "bulge out".

4. **Uncertainty in ΔB estimation:** Reading from the graph always has some error. However, using the paper's reported ΔB minimizes this as a source of discrepancy for this analysis.

5. **Gate effects / Depletion:** If there are nearby gates or charged surfaces, they could modify the effective conducting cross-section.

6. **Field penetration effects:** In some materials, the magnetic field inside the material might differ from the externally applied field. This is usually a minor effect for materials like (Bi,Sb)Te which are weakly magnetic.

The discrepancy is common in such experiments and is usually attributed to a combination of these factors, particularly the precise nature of electron paths and the actual morphology of the nanowire.